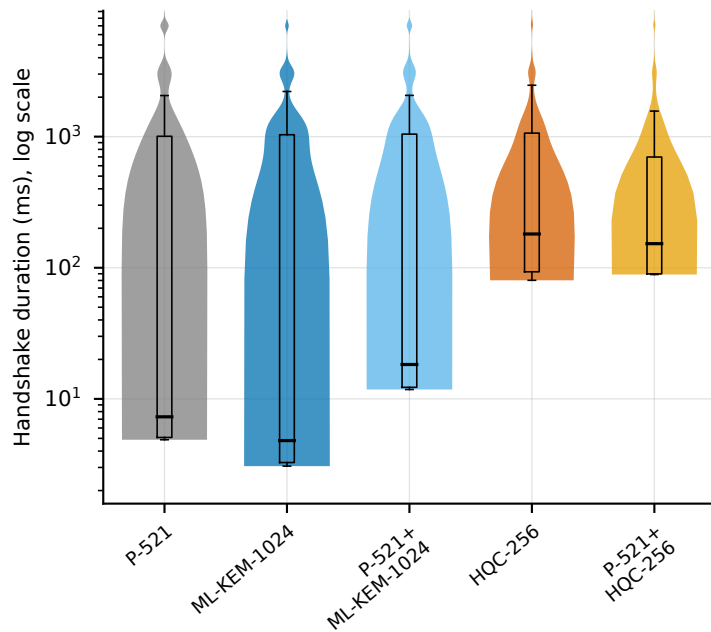
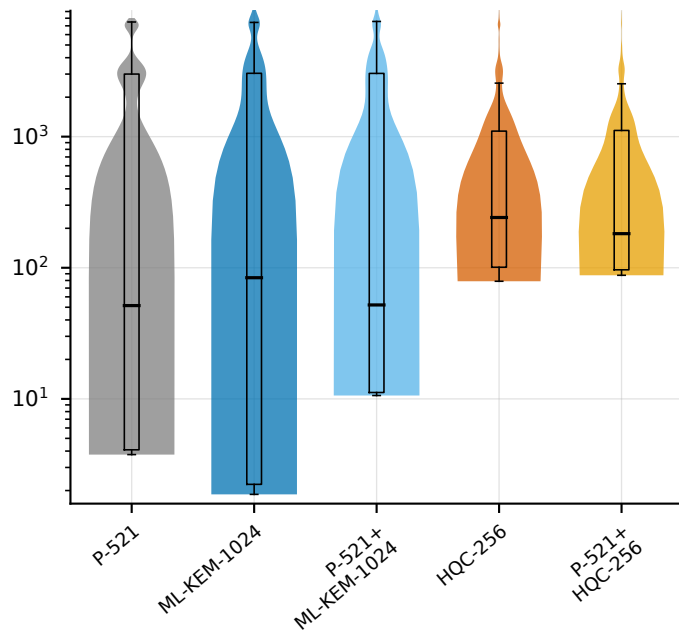


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	secp521r1 Fail	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)	mldsa87 Fail
P-521	1.03k	184.2	9.00k	28/500	1.51k	153.3	8.02k	48/500
ML-KEM-1024	846.1	168.0	7.85k	9/500	1.83k	147.9	15.99k	25/500
P-521+ML-KEM-1024	922.6	173.3	8.10k	6/500	1.72k	155.2	18.05k	20/500
HQC-256	785.0	171.1	9.25k	5/500	903.2	161.6	9.31k	6/500
P-521+HQC-256	681.4	200.5	14.12k	4/500	805.1	159.6	10.06k	7/500